Mohamed Ramy

Full-Stack Developer

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Alexandria, Egypt

Summary

Full-Stack Developer focused on designing robust, scalable systems end-to-end – from data models and APIs to the interfaces built on top of them. Experienced across the stack with **NestJS, Node.js, and PostgreSQL** on the backend and **Next.js** on the frontend, translating complex operational workflows into clean, maintainable products. Approaches every project by first understanding the problem at hand, then designing a system built to scale with it.

Technical Skills

Frontend: Next.js, React, TypeScript, Tailwind CSS, Responsive UI Design

Backend & Frameworks: NestJS, Node.js, Express.js, RESTful APIs, WebSockets (Socket.io), Microservices Architecture

Databases & ORM: PostgreSQL, MySQL, MongoDB (Vector Search), Prisma ORM, Mongoose, Redis

Security & Auth: JWT (JSON Web Tokens), Passport.js, Bcrypt, Role-Based Access Control (RBAC), OAuth2

Languages: TypeScript, JavaScript (ES6+), Python, C++ (Data Structures & Algorithms)

AI & Search: Google Gemini AI, RAG (Retrieval-Augmented Generation), ChromaDB, Prompt Engineering

Tools & DevOps: Git, GitHub, Jira, Docker (Basic), Postman, Insomnia, Linux

Projects

IA Platform – Education & Examination Management System Live Demo

Tech stack: NestJS, Next.js, PostgreSQL, Prisma, VdoCipher, Bunny.net

- **End-to-End Platform:** Designed and built both the Next.js frontend and NestJS backend for a centralized course, examination, and certification management system.
- **Examination Engine:** Engineered a configurable Computer-Based Testing system simulating national engineering exams, with timed sessions, question banks, automated grading, and performance analysis.
- **Role-Based Access Control:** Implemented role-based authorization for Super Admin, Admin, and Trainee accounts, enforcing access boundaries across administrative, course, and exam workflows.
- **Business Intelligence Dashboard:** Built an admin-facing analytics dashboard covering revenue, enrollment growth, course pass rates, and question-level failure rates for data-driven decision-making.
- **Content Security:** Integrated VdoCipher for protected video delivery and Bunny.net for controlled document hosting, restricting course materials to authorized trainees.

Mindset – AI-Powered RAG Platform GitHub

Tech stack: Node.js, MongoDB, Gemini AI, Vector Search

- **AI Architecture:** Implemented a full RAG pipeline using Google Gemini AI and MongoDB Vector Search for semantic retrieval from uploaded PDF documents.
- **API Design:** Developed an Express.js backend managing file processing, embedding generation, and contextual Q&A streaming, with a client interface for document upload and chat-based querying.

Experience

IEEE Computer Society – E-JUST Chapter Alexandria, Egypt
Webmaster Officer (Volunteer) *Oct 2024 – Present*

- **Full-Stack Maintenance**: Oversee frontend updates and API integrations for the official chapter website, ensuring 99.9% uptime for member registration services.
- **System Architecture**: Refactored backend logic to improve data retrieval speeds and support new feature sets for event management.

Freelance Academic Tutoring Alexandria, Egypt
Data Structures & Algorithms Instructor *Sep 2023 – Jun 2024*

- **Core CS Instruction**: Taught **Data Structures and Algorithms in C++** to 100+ students, focusing on time/space complexity analysis and efficient memory management.
- **Technical Mentorship**: Developed custom coding challenges focused on algorithmic problem-solving, a cornerstone of well-rounded software engineering.

Education

Egypt-Japan University of Science and Technology (E-JUST) Alexandria, Egypt
B.Sc. in Computer Science and Information Technology — GPA: 3.3/4.0 *Sep 2023 – Jun 2027 (Expected)*

Honors & Awards

EJUST AI-Accelerated Appathon – 1st Place Alexandria, Egypt
Prize: EGP 15,000 *2025*

- **Achievement**: Secured 1st place among 20+ teams for the technical architecture of an AI-driven educational platform.